

Cloud LearnX

Mathematics

Transformations — Practice Worksheet

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Topic: Transformations

Instructions:

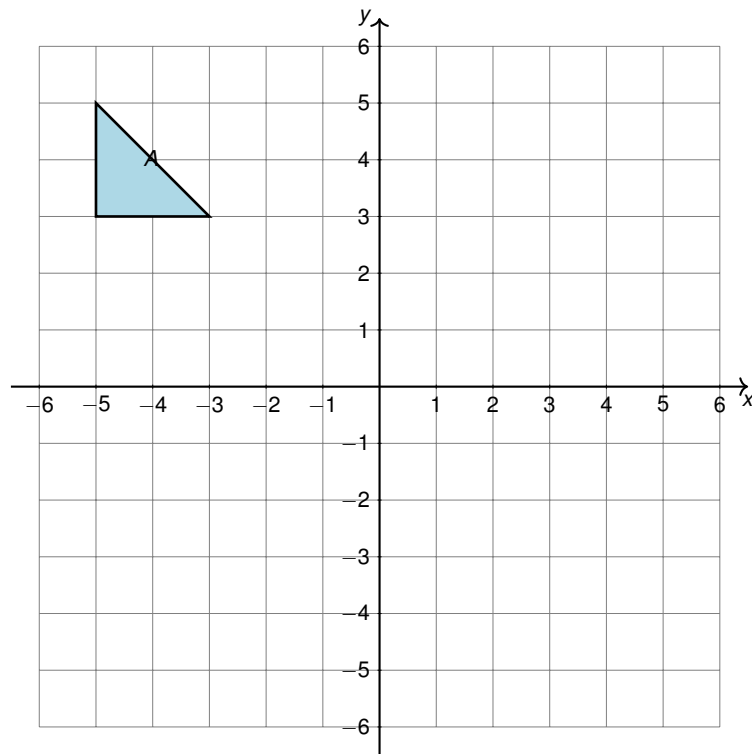
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** In a Valorant tournament, the map designer represents the positions of two objective markers, triangle A , on a coordinate grid. Triangle A is translated by the vector $\begin{pmatrix} 4 \\ -5 \end{pmatrix}$ to give triangle B .

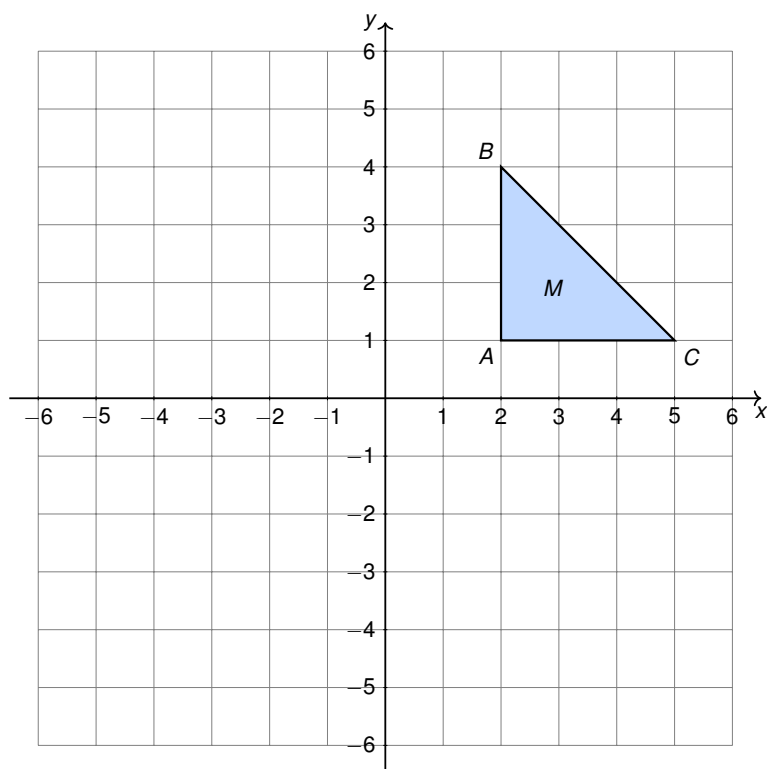


On the grid, draw triangle B , the image of triangle A after this translation.

(2)

(Total for Question 1 is 2 marks)

- 2 A game designer is creating a symmetrical map layout for a Valorant-style level. Triangle M marks the position of one team's spawn zone, as shown on the grid below.

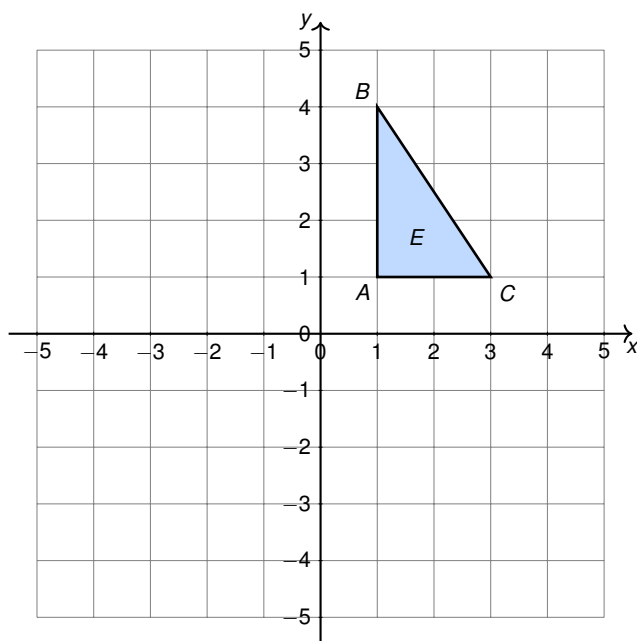


Reflect triangle M in the x -axis.
Label the image N .

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(2)

(Total for Question 2 is 2 marks)

3 Triangle E has vertices at $(1, 1)$, $(1, 4)$ and $(3, 1)$, as shown on the grid below.



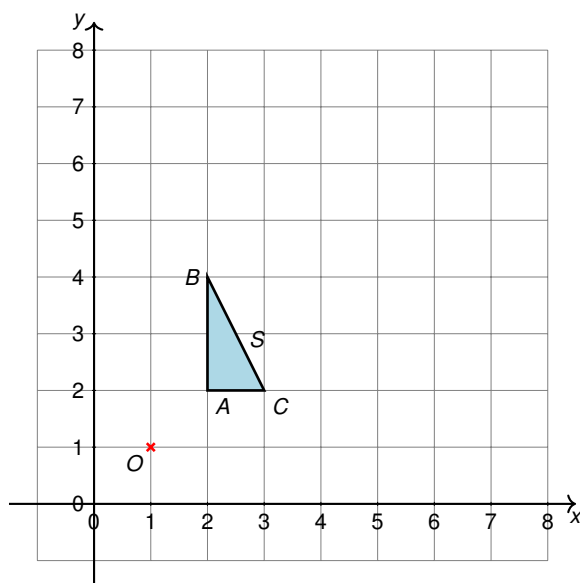
Rotate triangle E by 90° clockwise about the origin.

Draw and label the image triangle E' on the grid, showing the coordinates of each vertex.

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(2)

(Total for Question 3 is 2 marks)

4 Triangle S is shown on the grid below.

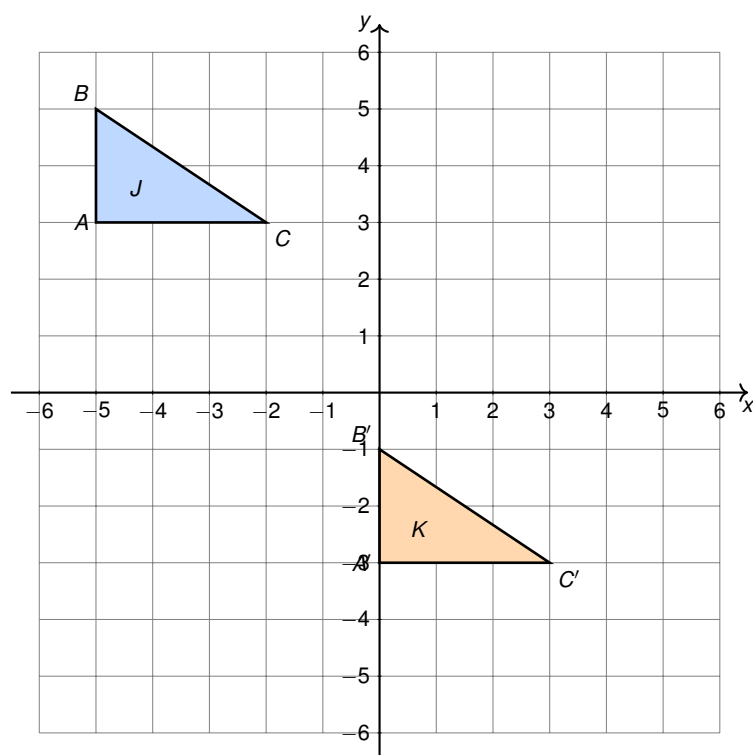


Enlarge triangle S by scale factor 2, using O as the centre of enlargement.
Label the image S' .

(3)

(Total for Question 4 is 3 marks)

- 5 On a Valorant esports tournament overlay, a designer uses a coordinate grid to animate a map icon. Triangle J is translated to give triangle K , as shown on the grid below.



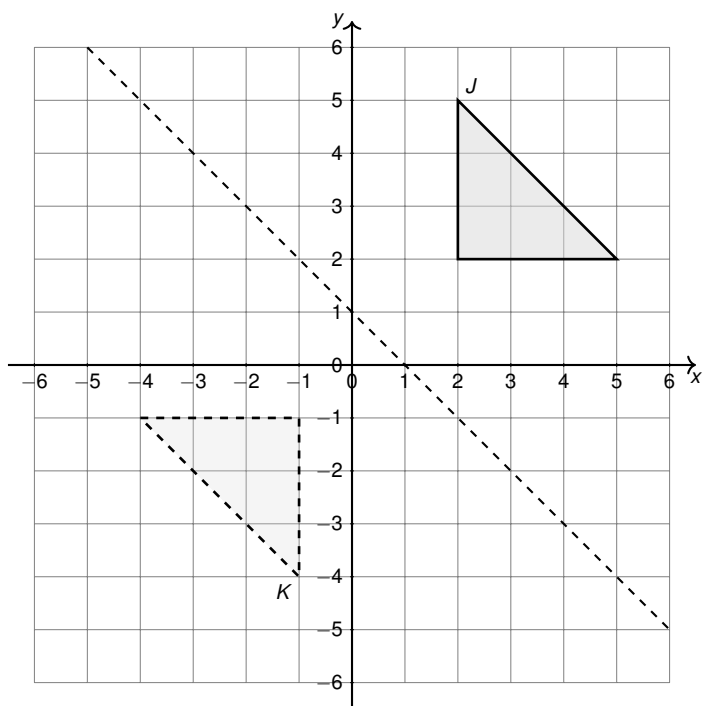
Describe fully the translation that maps triangle J onto triangle K .

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(2)

(Total for Question 5 is 2 marks)

- 6 On a Valorant esports tournament overlay, a designer plots the positions of two identical map icons, triangle J and triangle K , on a coordinate grid to test a mirroring animation effect.



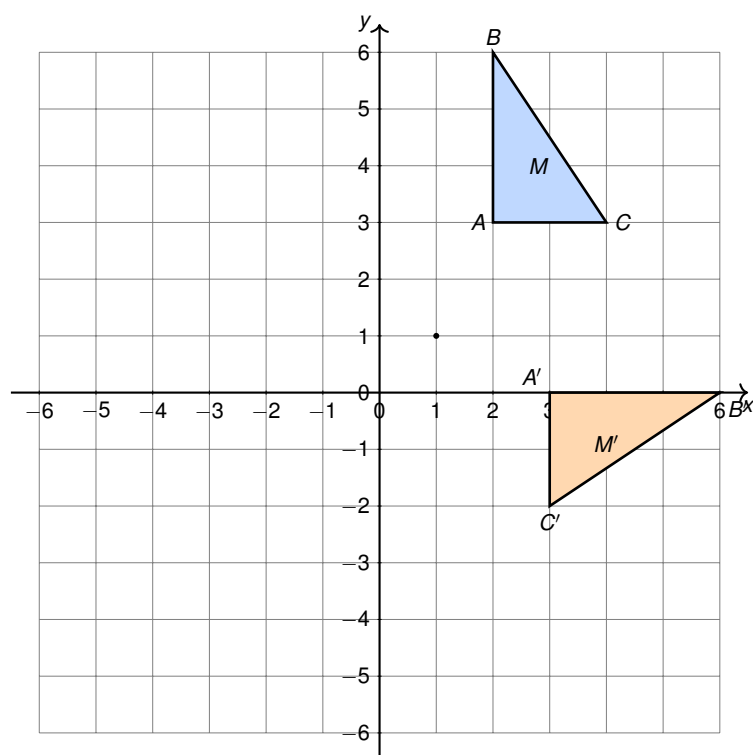
Triangle J is reflected onto triangle K , as shown on the grid.

Describe fully the single transformation that maps triangle J onto triangle K .

(2)

(Total for Question 6 is 2 marks)

- 7 An esports tournament designer is creating a symmetric logo for a Valorant-themed banner. Triangle M is rotated to give triangle M' , as shown on the grid.

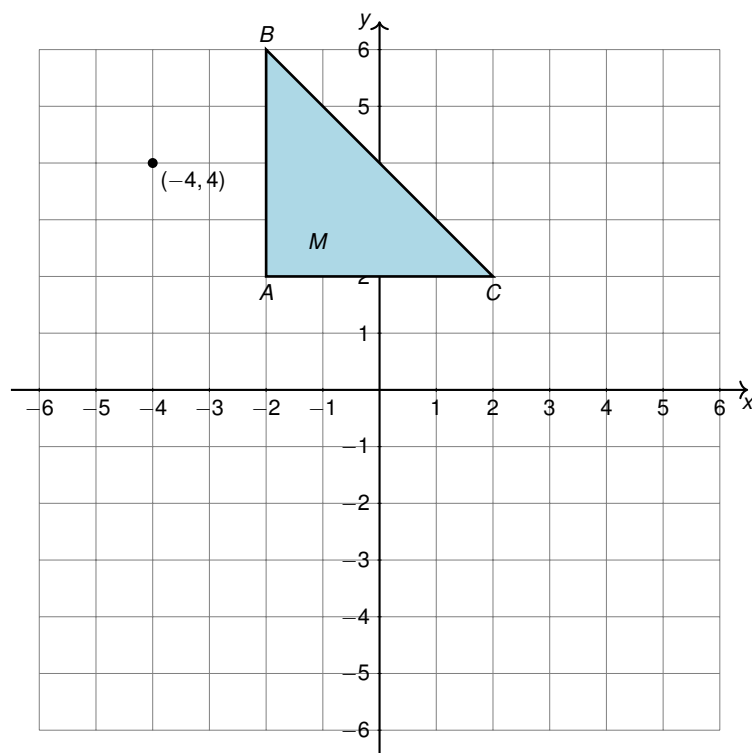


Describe fully the single transformation that maps triangle M onto triangle M' .

(3)

(Total for Question 7 is 3 marks)

- 8 Shape M is enlarged by scale factor $\frac{1}{2}$, using centre of enlargement $(-4, 4)$, to give shape M' .



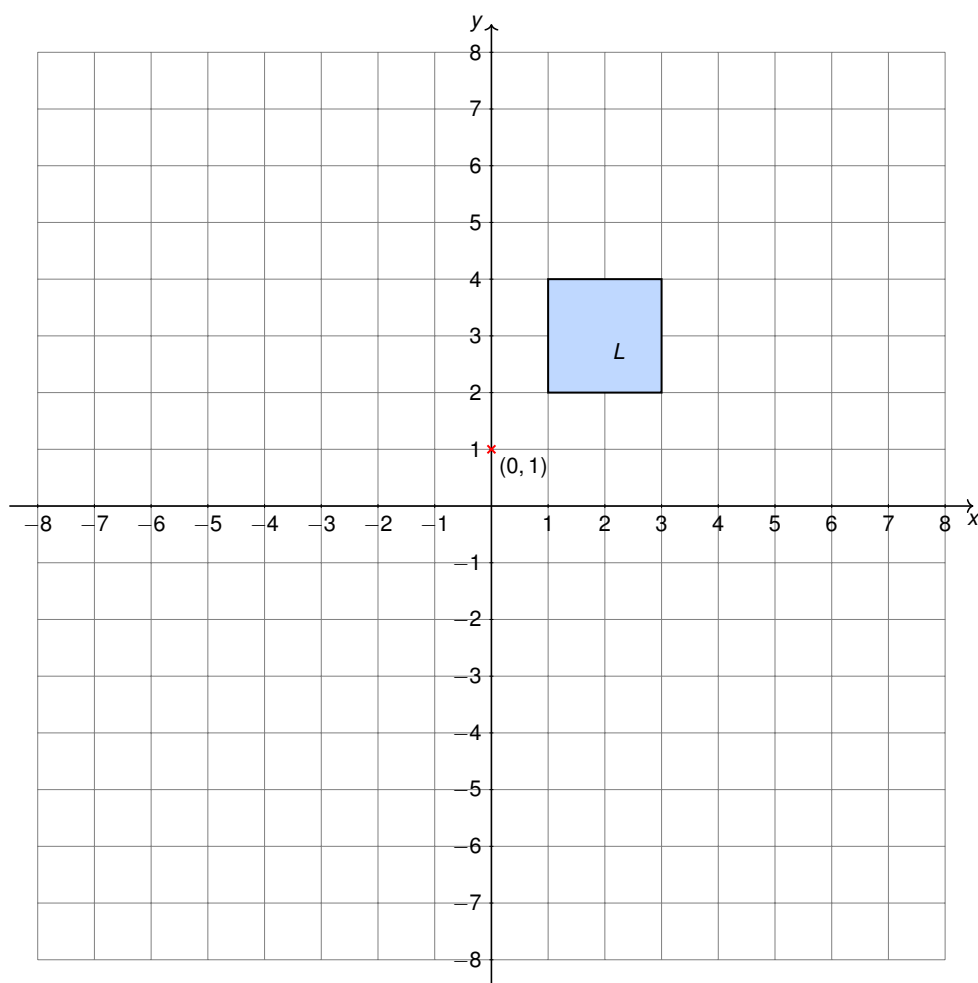
On a copy of the grid, draw the image of shape M after the enlargement, and label it M' .
State the coordinates of the vertices of M' .

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(3)

(Total for Question 8 is 3 marks)

- 9 Shape L is drawn on the grid below. A designer wants to create a logo L' by enlarging L with scale factor -2 , using centre of enlargement $(0, 1)$.



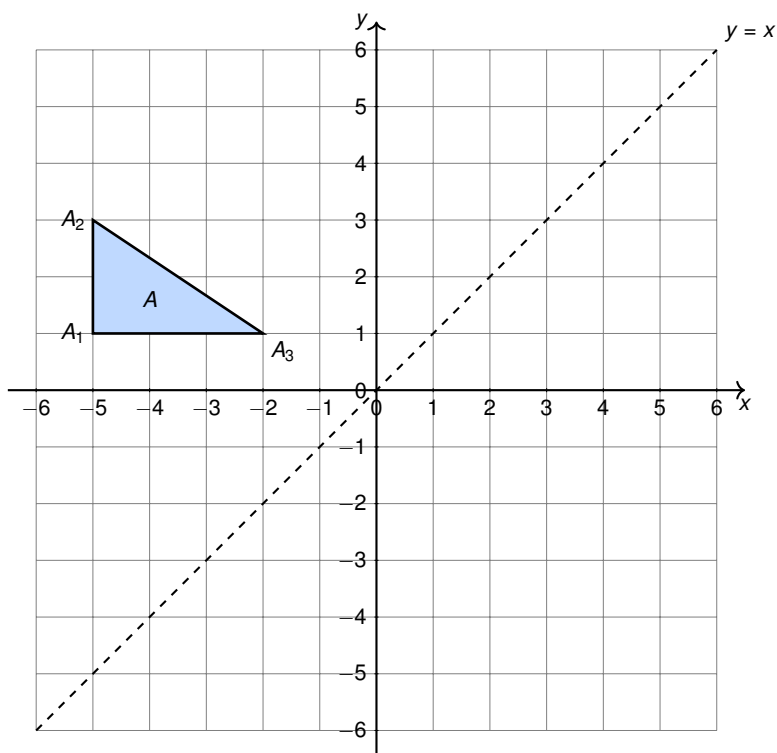
The vertices of shape L are $A(1, 2)$, $B(1, 4)$, $C(3, 4)$ and $D(3, 2)$.

On squared paper, the coordinates of shape L have been given above (no grid drawing required for your answer). Work out the coordinates of the vertices A' , B' , C' and D' of the image L' after this enlargement.

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(4)

(Total for Question 9 is 4 marks)

- 10** On a coordinate plane used to design an e-sports tournament bracket display, triangle A has vertices $A_1(-5, 1)$, $A_2(-5, 3)$, $A_3(-2, 1)$.



- (a) Triangle A is translated by the vector $\begin{pmatrix} 4 \\ -5 \end{pmatrix}$ to give triangle B .
Find the coordinates of the vertices of triangle B .

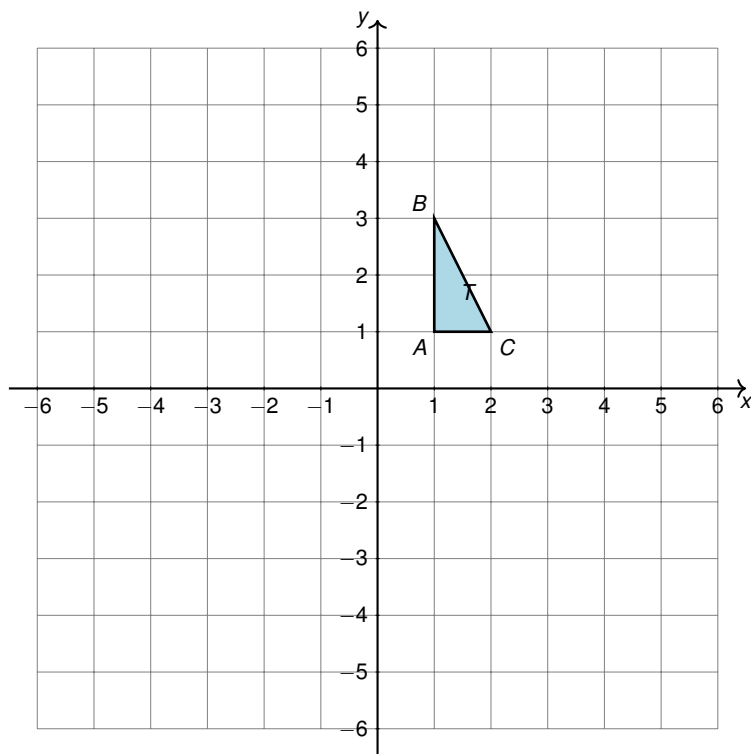
.....
(2)

- (b) Triangle B is then reflected in the line $y = x$ to give triangle C .
Find the coordinates of the vertices of triangle C .

.....
(2)

(Total for Question 10 is 4 marks)

11 Triangle T has vertices $A(1, 1)$, $B(1, 3)$ and $C(2, 1)$.



Triangle T is first rotated 90° clockwise about the origin $(0, 0)$ to give triangle T_1 .
Triangle T_1 is then reflected in the x -axis to give triangle T_2 .

(a) Write down the coordinates of the vertices of triangle T_2 .

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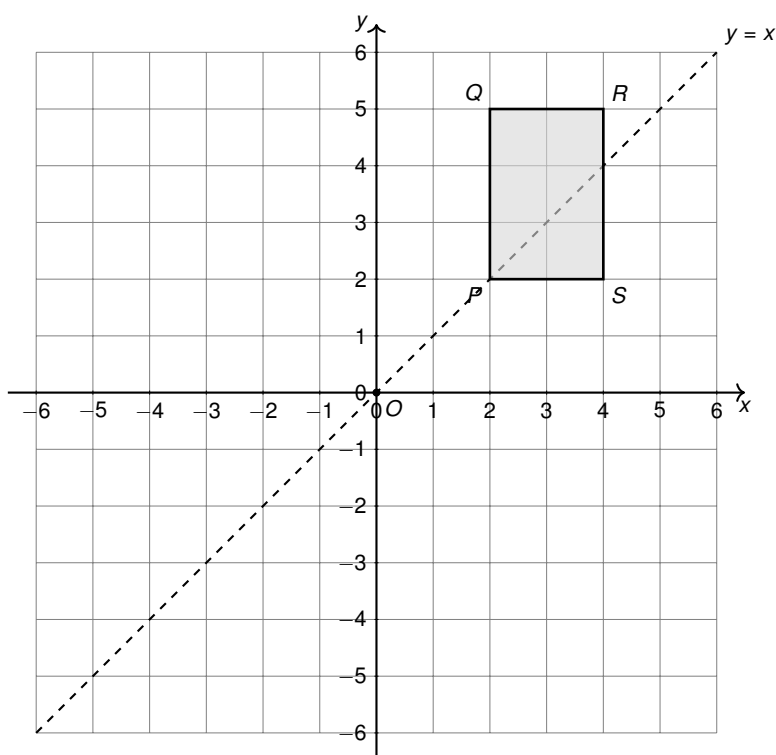
(2)

(b) Describe **fully** the **single** transformation that maps triangle T directly onto triangle T_2 .
(A description involving two transformations will not gain full credit.)

(2)

(Total for Question 11 is 4 marks)

- 12 A games map designer is testing symmetry rules for a Valorant-style tactical map editor. A rectangular control zone $PQRS$ is drawn on the design grid, together with the line $y = x$ and the origin O .



The rectangle $PQRS$ has vertices $P(2, 2)$, $Q(2, 5)$, $R(4, 5)$ and $S(4, 2)$.

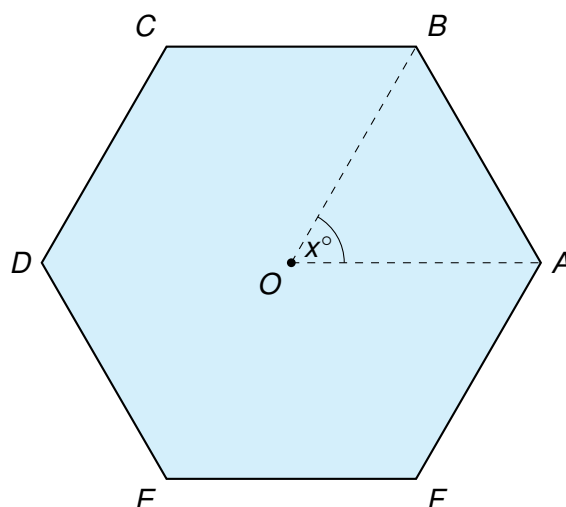
The rectangle is first reflected in the line $y = x$ to give rectangle $P_1Q_1R_1S_1$.

Rectangle $P_1Q_1R_1S_1$ is then rotated 180° about the origin O to give rectangle $P_2Q_2R_2S_2$.

By considering the effect of **both** transformations on each vertex of $PQRS$, find the coordinates of any point(s) on the perimeter of the rectangle that are invariant under the **combined** transformation (reflection in $y = x$ followed by rotation 180° about O).

(Total for Question 12 is 3 marks)

- 13 An e-sports arena has a rotating stage logo shaped like a regular hexagon, used as the map-select icon in a Valorant tournament broadcast. The graphic designer draws the hexagonal logo on a coordinate grid centred at the origin O , as shown below.



The logo $ABCDEF$ is a regular hexagon with centre O . During the broadcast animation, the logo is rotated repeatedly about O so that it always lands back exactly on top of itself (vertex meeting vertex).

- (a) Work out the smallest angle x , marked on the diagram, through which the logo must be rotated about O so that it maps onto itself.

$$x = \dots\dots\dots^\circ$$

(1)

- (b) State the order of rotational symmetry of the hexagonal logo.

\dots\dots\dots

(1)

- (c) The animation designer instead wants the logo to complete a sequence of rotations that brings vertex A to the position currently occupied by vertex D , using the **smallest possible number of equal individual rotation steps** (each step being one of the symmetry rotations found in part (a)), and each step must be in the **same direction**. Work out the number of individual rotation steps needed, and state the direction (clockwise or anticlockwise) that requires the fewest steps.

\dots\dots\dots

(1)

(Total for Question 13 is 3 marks)

- 14** In an e-sports arena, a robotic camera drone glides across a coordinate grid (units in metres) to capture footage of a Valorant tournament. The drone starts at position $A(-4, 5)$, hovering above one team's spawn point, and moves in a straight line to position $B(6, -3)$, above the opposing team's spawn point.

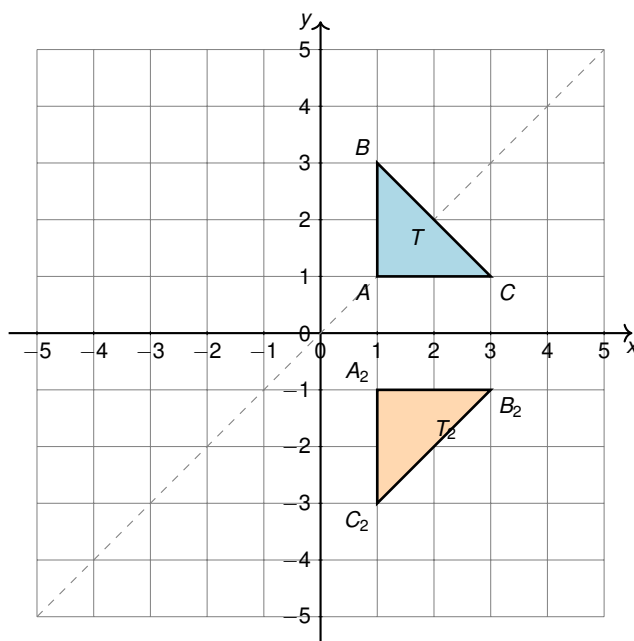
Find the column vector that describes the translation which takes the drone from A to B .

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(3)

(Total for Question 14 is 3 marks)

- 15** In a Valorant map-design tool, a designer places a triangular "spike site" marker T on a coordinate grid, with vertices $A(1, 1)$, $B(1, 3)$ and $C(3, 1)$.

The designer first reflects T in the line $y = x$ to get triangle T_1 . She then reflects T_1 in the x -axis to get triangle T_2 , as shown on the grid below.



- (a) Find the coordinates of the vertices of T_1 (the image of T after reflecting in the line $y = x$), and hence the coordinates of the vertices of T_2 (the image of T_1 after reflecting in the x -axis).

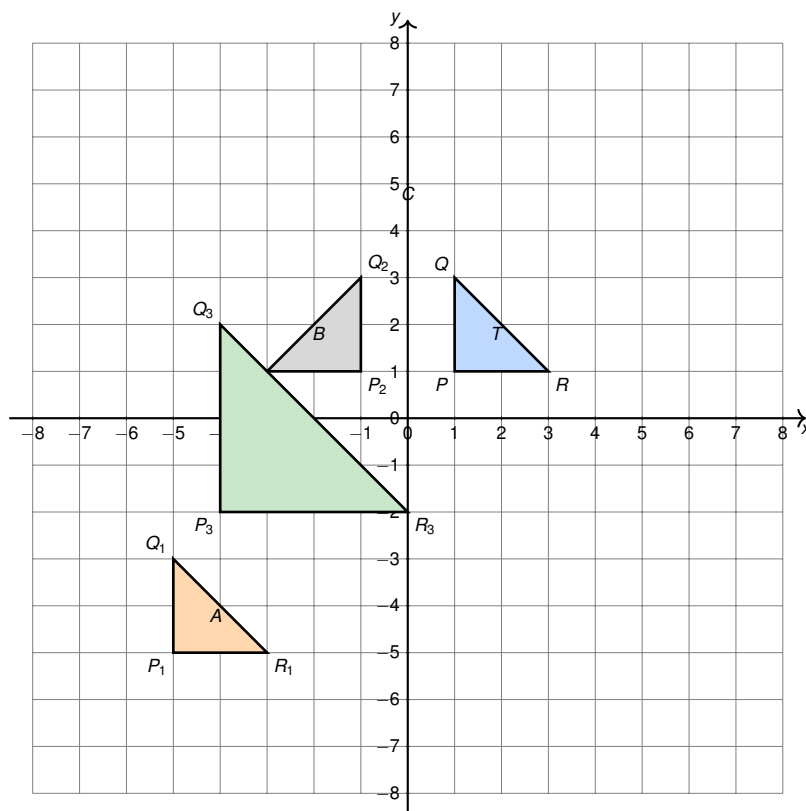
(2)

- (b) By comparing the coordinates of T and T_2 , show that the combined transformation (reflection in $y = x$ followed by reflection in the x -axis) is equivalent to a single rotation, stating the angle, direction and centre of this rotation.

(2)

(Total for Question 15 is 4 marks)

16 The diagram shows triangle T and its images A , B and C after three different single transformations.



Triangle T has vertices $P(1, 1)$, $Q(1, 3)$ and $R(3, 1)$.

- (a) Triangle A (vertices P_1, Q_1, R_1) is the image of T under a single transformation. Describe this transformation fully.

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- (b) Triangle B (vertices P_2, Q_2, R_2) is the image of T under a single transformation. Describe this transformation fully.

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(2)

- (c) Triangle C (vertices P_3, Q_3, R_3) is the image of T under a single transformation. Describe this transformation fully, stating the coordinates of the centre.

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(1)

(Total for Question 16 is 5 marks)

TOTAL FOR PAPER IS ?? MARKS